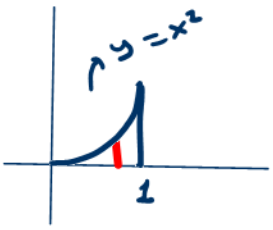
Cevaplar

① $\int_0^1 \int_{\sqrt{y}}^1 \frac{dx\,dy}{\sqrt{1+3x^3}}$ integrasyon sırasını değiştirirsek

D: $y=1$ $y=0$ → x-eks.
$x=\sqrt{y}$ $x=1$
↓
$y=x^2$



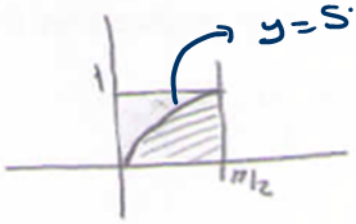
$I = \int_0^1 \int_0^{x^2} \frac{dy\,dx}{\sqrt{1+3x^3}}$

Cevap B

---

② $\int_0^1 \int_{\text{Arcsiny}}^{\pi/2} e^{\cos x}\,dx\,dy$ int. sırasını değiştirirsek yeni integral?

D: $y=1$ $y=0$
$x=\frac{\pi}{2}$ $x=\text{ArcSiny}$
$y=\text{Sinx}$



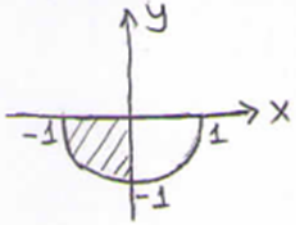
$I = \iint_D e^{\cos x}\,dy\,dx = \int_0^{\pi/2} \int_0^{\text{Sinx}} e^{\cos x}\,dy\,dx$

Cevap C

---

③ $\int_{-1}^0 \int_{-\sqrt{1-x^2}}^0 \frac{2}{1+\sqrt{x^2+y^2}}\,dy\,dx$ kutupsal dönüşüm ile yazınca oluşan integral?

R: $\begin{cases} x=0\,,\ x=-1 \\ y=0\,,\ y=-\sqrt{1-x^2} \end{cases}$



$I = \int_{\pi}^{\frac{3\pi}{2}} \int_0^1 \frac{2}{1+\underbrace{\sqrt{r^2}}_{r}}\,r\,dr\,d\theta$

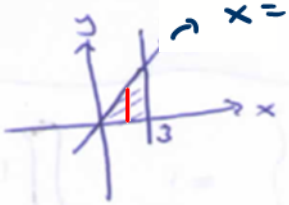
Cevap B

④ $\int_0^1 \int_{3y}^{3} e^{x^2} dx dy$ int. sırasını değiştirince yeni integral?

$x = 3y \rightarrow y = \frac{x}{3}$

x-ekseni

D: $y=0$ $y=1$ $x=3$ $x=3y$



$$I = \int_0^3 \int_0^{x/3} e^{x^2} dy dx$$

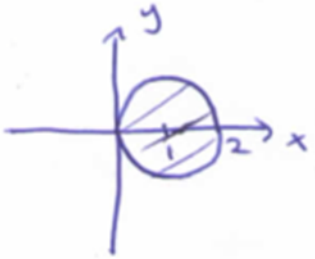
Cevap B

---

⑤ D: $x^2+y^2=2x$ bölgesinde $f(x,y)=x^2+y^2$ fonksiyonunun iki katlı integralini kutupsal koor. yazınız.

a) $\int_{-\pi/2}^{\pi/2} \int_0^1 r^3 dr d\theta$ b) $\int_{-\pi/2}^{\pi/2} \int_0^2 r^3 dr d\theta$ c) $\int_{-\pi/2}^{\pi/2} \int_0^{2\cos\theta} r^3 dr d\theta$

$x^2+y^2-2x=0 \Rightarrow (x-1)^2+y^2=1$



$x = r\cos\theta$
$y = r\sin\theta$
$dxdy = r\,dr\,d\theta$
$x^2+y^2=r^2$

$$I = \int_{-\pi/2}^{\pi/2} \int_0^{2\cos\theta} r^2 \cdot r\, dr\, d\theta$$

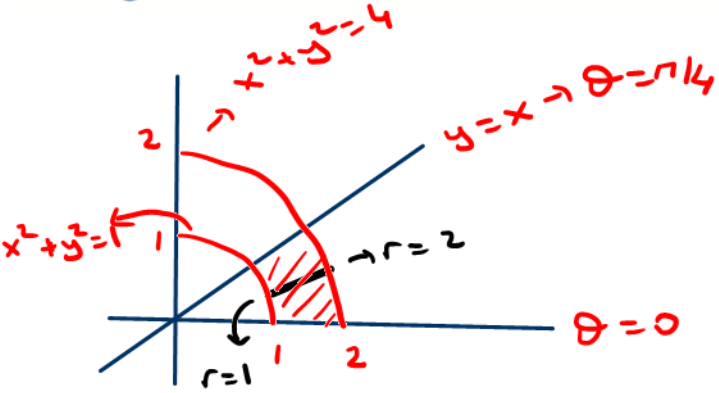
Cevap C

$x^2+y^2-2x=0$

$r^2-2r\cos\theta=0$ $r=0$ $r=2\cos\theta$

⑥ R bölgesi : $\{(x,y)\in\mathbb{R}^2 \mid 1\le x^2+y^2\le 4,\ 0\le y\le x\}$ olmak üzere $\iint_R \text{Arctan}\left(\frac{y}{x}\right)dA$ integralini kutupsal koor. yazınız.

a) $\int_0^{\pi/2}\int_1^4 \theta\, dr\,d\theta$ b) $\int_0^{\pi/4}\int_1^2 \theta\, r\, dr\, d\theta$ c) $\int_0^{\pi/4}\int_1^2 r\,\text{Tan}\theta\, dr\, d\theta$



$$I=\int_0^{\pi/4}\int_1^2 \theta\cdot r\cdot dr\,d\theta$$

Cevap B

$$\left.\begin{matrix} x=r\cos\theta \\ y=r\sin\theta \end{matrix}\right\} \frac{y}{x}=\tan\theta$$

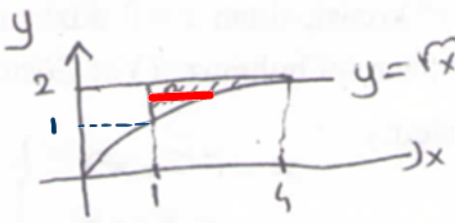
$dx\,dy=r\,dr\,d\theta$

$\text{Arctan}\frac{y}{x}=\text{Arctan}(\tan\theta)=\theta$

---

⑦ $\int_1^4\int_{\sqrt{x}}^2 \frac{e^y}{y+1}\,dy\,dx$ int. sırasını değiştirince yeni int.?

R: $\begin{cases} y=\sqrt{x},\ y=2 \\ x=1,\ x=4 \end{cases}$

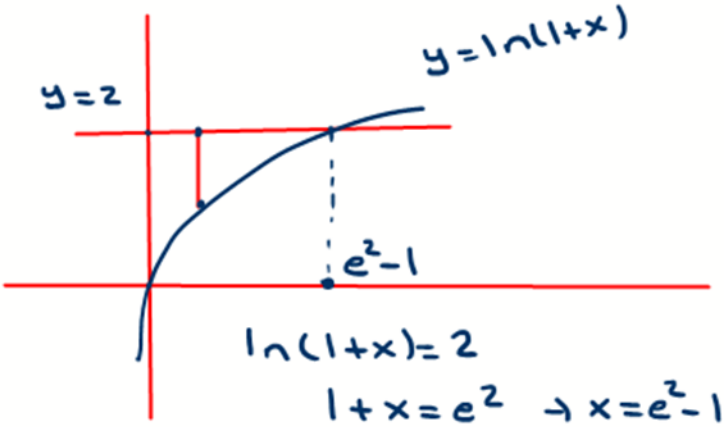


$$I=\int_1^2\int_1^{y^2}\frac{e^y}{y+1}\,dx\,dy$$

Cevap B

(8) $y=\ln(1+x)$, $x=0$, $y=2$ ile sınırlı bölgenin alanını veren integral hangisi(leri) olabilir?

I. $\int_0^2 \int_0^{e^y} dx\,dy$ II. $\int_0^{e^2-1} \int_{\ln(1+x)}^{2} dy\,dx$ III. $\int_0^2 \int_0^{e^y-1} dx\,dy$

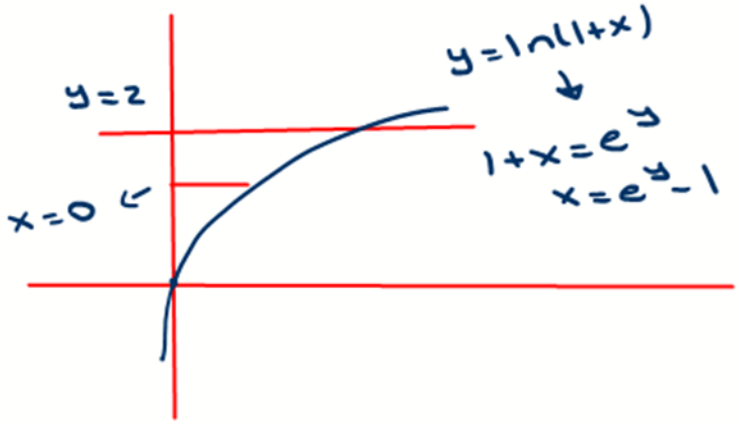


$\ln(1+x)=2$

$1+x=e^2 \rightarrow x=e^2-1$

★ x'e göre

$\int_0^{e^2-1} \int_{\ln(1+x)}^{2} dy\,dx$ → (II)



★ y'ye göre

$\int_0^{2} \int_0^{e^y-1} dx\,dy$ → (III)

Cevap C

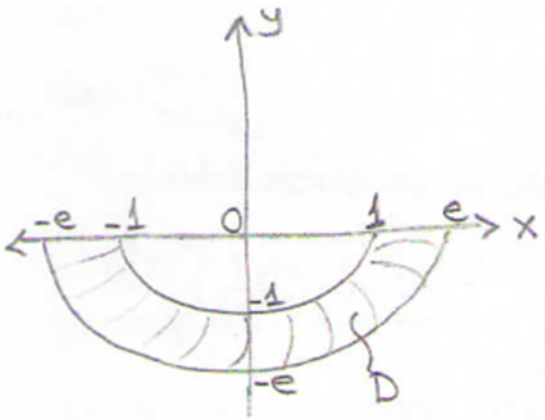
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(9) $D=\{(x,y)\in\mathbb{R}^2 \mid 1\le x^2+y^2\le e^2 , y\le 0\}$ bölgesinin alanını veren kutupsal integral?

$x=r\cos\theta$

$y=r\sin\theta$

$dx\,dy=r\,dr\,d\theta$



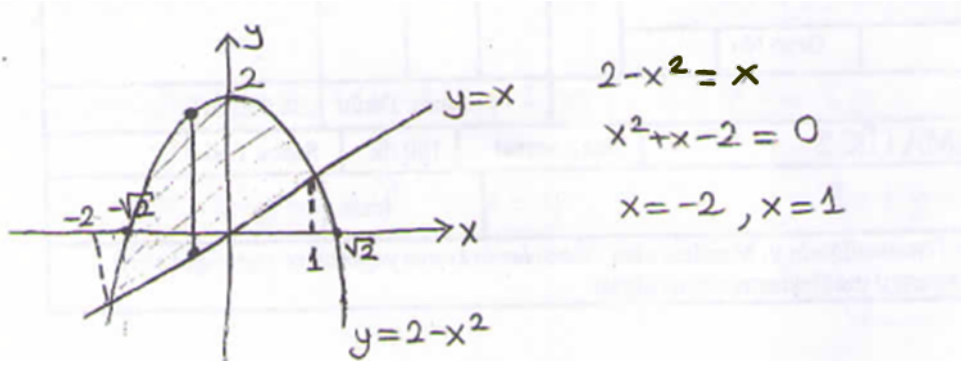
$A=\iint_D dx\,dy$

$=\int_{\pi}^{2\pi} \int_1^e r\,dr\,d\theta \quad r\,d\theta$

Cevap C

(10) D bölgesi $y = 2-x^2$ ve $y = x$ ile sınırlı bölge olsun. $z = 2x^2y$ ile üstten, $z=0$ daki D bölgesi ile alttan sınırlanan cismin hacmini veren integral?

a) $\int_0^1 \int_x^{2-x^2} 2x^2y \, dy\,dx$ b) $\int_{-2}^1 \int_x^{2-x^2} 2x^2 \, dy\,dx$ c) $\int_{-2}^2 \int_0^{2-x^2} 2x^2y \, dy\,dx$



$2-x^2 = x$

$x^2+x-2 = 0$

$x=-2, \; x=1$

$V = \iint_D 2x^2y \, dx\,dy$

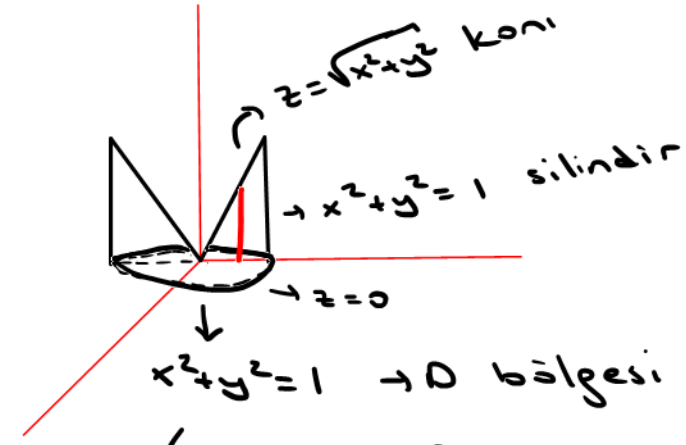
$= \int_{-2}^1 \int_x^{2-x^2} 2x^2y \, dy\,dx$

Cevap B

---

(11) üstten $z=\sqrt{x^2+y^2}$ yüzeyi, alttan $z=0$ düzlemi, yandan $x^2+y^2=1$ yüzeyi ile sınırlı cismin hacmini veren integral?

a) $\int_0^{2\pi} \int_0^1 r \, dr\,d\theta$ b) $\int_0^{2\pi} \int_0^1 r^2 \, dr\,d\theta$ c) $\int_0^{\pi} \int_0^1 r^2 \, dr\,d\theta$



$x = r\cos\theta$

$y = r\sin\theta$

$x^2+y^2 = r^2$

$dx\,dy = r\,dr\,d\theta$

$V = \iint_D \underbrace{(\text{yükseklik})}_{\sqrt{x^2+y^2}-0} \, dx\,dy$

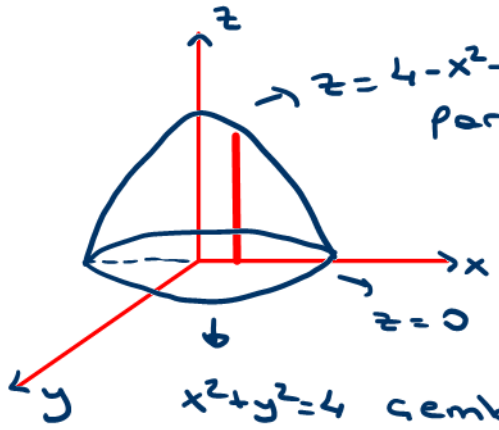
$= \iint_D \sqrt{x^2+y^2} \, dx\,dy$

$= \int_0^{2\pi} \int_0^1 r \cdot r \, dr\,d\theta$

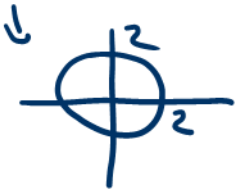
Cevap B

(12) $z = 4 - x^2 - y^2$ ve $z = 0$ yüzeylerinin sınırladığı cismin hacmini veren int.?

a) $\int_0^{2\pi} \int_0^4 r\,dr\,d\theta$ b) $\int_0^{2\pi} \int_0^2 (4r - r^3)\,dr\,d\theta$ c) $\int_0^{2\pi} \int_0^1 (4r - r^3)\,dr\,d\theta$



$x^2 + y^2 = 4$ çemberi D bölgesi



$x = r\cos\theta$
$y = r\sin\theta$
$x^2 + y^2 = r^2$
$dx\,dy = r\,dr\,d\theta$

$$V = \iint_D (4 - \text{üst}) - \text{alt})\,dx\,dy$$

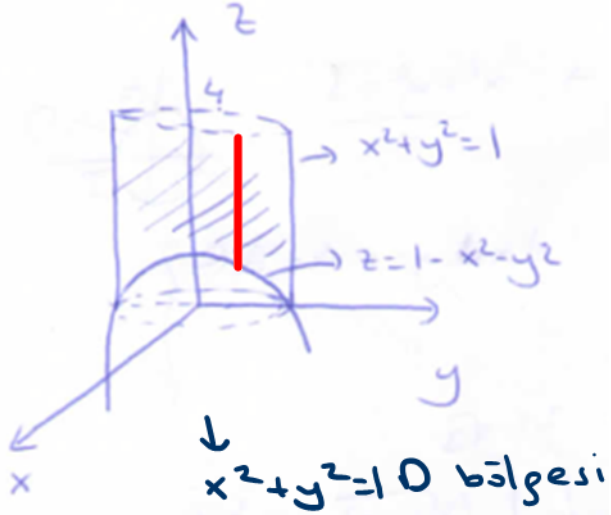
$$= \iint_D (4 - x^2 - y^2)\,dx\,dy$$

$$= \int_0^{2\pi} \int_0^2 (4 - r^2) r\,dr\,d\theta$$

Cevap B

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(13) $x^2 + y^2 = 1$, $z = 4$, $z = 1 - x^2 - y^2$ yüzeyleri ile sınırlı cismin hacmini veren integral?



$x^2 + y^2 = 1$ D bölgesi



$x^2 + y^2 = r^2$
$dx\,dy = r\,dr\,d\theta$
$x = r\cos\theta$
$y = r\sin\theta$

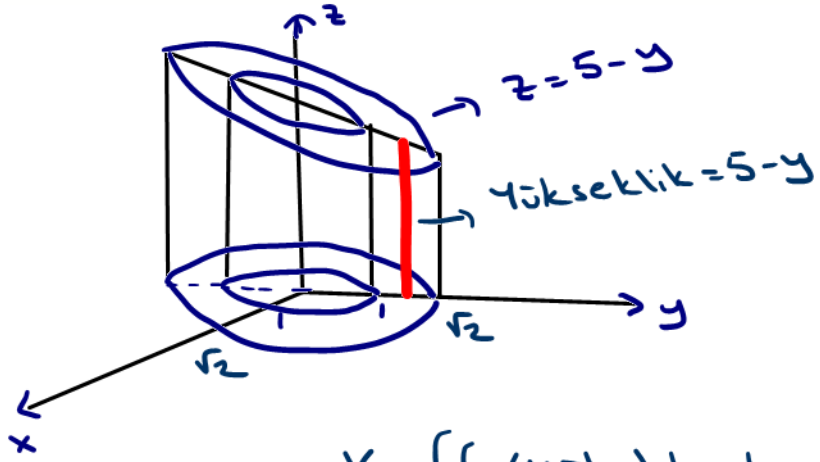
$$V = \iint_D \underbrace{(\text{üst} - \text{alt})}_{4 - (1 - x^2 - y^2)}\,dx\,dy$$

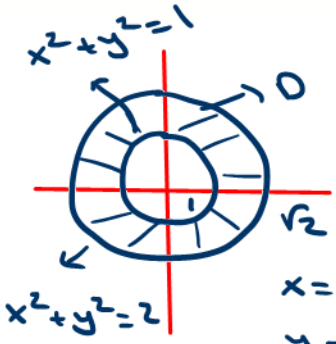
$$= \iint_D (3 + x^2 + y^2)\,dx\,dy$$

$$= \int_0^{2\pi} \int_0^1 (3 + r^2) r\,dr\,d\theta$$

Cevap C

(14) $5=z+y$ düzleminin altında, $z=0$ düzleminin üstünde, $x^2+y^2=2$ ve $x^2+y^2=1$ silindirleri arasında kalan cismin hacmini veren int.?





$x=r\cos\theta$

$y=r\sin\theta$

$dxdy=rdrd\theta$

$$V=\iint_0 (\text{yük.})\,dxdy$$

$$=\iint_0 (5-y)\,dxdy=\int_0^{2\pi}\int_1^{\sqrt{2}} (5-r\sin\theta)\,r\,dr\,d\theta$$

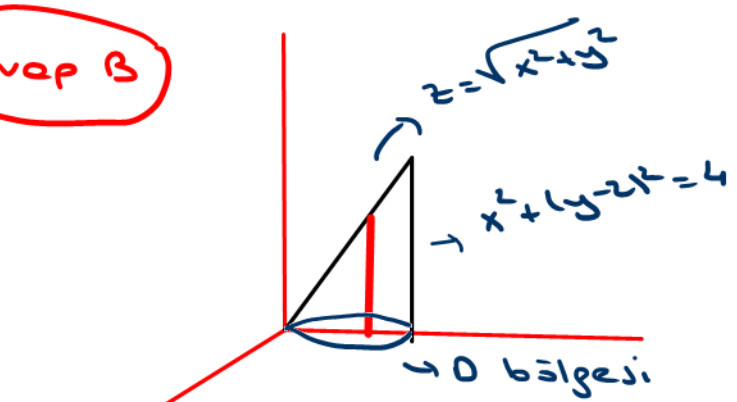
Cevap A

---

(15) $x^2+y^2-4y=0$, $z=0$, $z=\sqrt{x^2+y^2}$ yüzeyleri arasındaki cismin hacmini veren integral?

a) $\int_0^{2\pi}\int_0^{4} r^2\,dr\,d\theta$ b) $\int_0^{\pi}\int_0^{4\sin\theta} r^2\,dr\,d\theta$ c) $\int_0^{\pi}\int_0^{2} r^2\,dr\,d\theta$

Cevap B



(*) $x^2+y^2-4y+4-4=0$

$x^2+(y-2)^2=4$

ötelenmiş silindir

(x) $z=\sqrt{x^2+y^2}$ → Koni

$x^2+(y-2)^2=4 \to 0 \to x^2+y^2=4y \to r^2=4r\sin\theta$

$r=4\sin\theta$



$x=r\cos\theta$

$y=r\sin\theta$

$dxdy=rdrd\theta$

$$V=\iint_0 \sqrt{x^2+y^2}\,dxdy$$

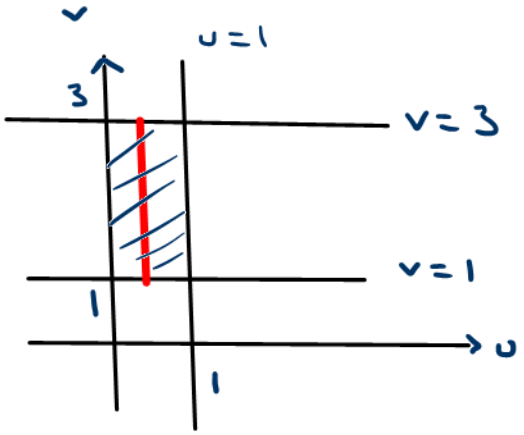
$$=\int_0^{\pi}\int_0^{4\sin\theta} r\cdot r\,dr\,d\theta$$

(16) $R:\begin{cases} x+y=1 \\ x+y=3 \\ x-y=0 \\ x-y=1 \end{cases}$ bölgesinde $I=\iint_R \frac{x^2-y^2}{\sqrt{1+(x-y)^2}}\,dA$ integralini $u=x-y$ $v=x+y$ dönüşümü ile yeniden yazınız.

$u=x-y \quad v=x+y$

①

R	R'
$x+y=1 \longrightarrow$	$v=1$
$x+y=3 \longrightarrow$	$v=3$
$x-y=0 \longrightarrow$	$u=0$
$x-y=1 \longrightarrow$	$u=1$



② $J=\dfrac{1}{\begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}}=\dfrac{1}{\begin{vmatrix} 1 & -1 \\ 1 & 1 \end{vmatrix}}=\dfrac{1}{2} \Rightarrow dx\,dy=\dfrac{1}{2}du\,dv$

③ $I=\iint_R \underbrace{\frac{x^2-y^2}{\sqrt{1+\underbrace{(x-y)^2}_{u^2}}}}_{u.v}\,\underbrace{dx\,dy}_{\frac{1}{2}du\,dv}=\int_0^1\int_1^3 \frac{uv}{\sqrt{1+u^2}}\cdot\frac{1}{2}\,dv\,du$

Cevap A

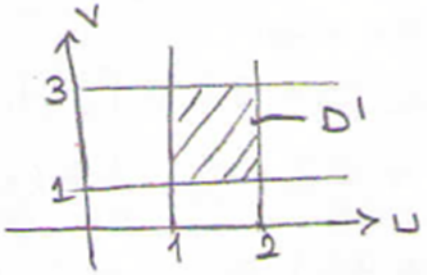
(17) $D:\begin{cases} y=x^2 \\ y=2x^2 \\ x=y^2 \\ x=3y^2 \end{cases}$ bölgesinin alanını $u=\frac{y}{x^2}$ $v=\frac{x}{y^2}$ dönüşümü ile yazarsak integrali yeniden integral?

a) $\int_1^2\int_1^3 \frac{1}{3u^2v^2}\,dv\,du$ b) $\int_1^2\int_1^3 dv\,du$ c) $\int_1^3\int_1^3 \frac{1}{3u^2v^2}\,du\,dv$

$u=\frac{y}{x^2}$

$v=\frac{x}{y^2}$

D	D'
$\frac{y}{x^2}=1 \rightarrow$	$u=1$
$\frac{y}{x^2}=2 \rightarrow$	$u=2$
$\frac{x}{y^2}=1 \rightarrow$	$v=1$
$\frac{x}{y^2}=3 \rightarrow$	$v=3$



$$\frac{D(x,y)}{D(u,v)}=\frac{1}{\frac{D(u,v)}{D(x,y)}}=\frac{1}{\begin{vmatrix} -2y/x^3 & 1/x^2 \\ 1/y^2 & -2x/y^3 \end{vmatrix}}=\frac{x^2y^2}{3}=\frac{1}{3u^2v^2}$$

Cevap A

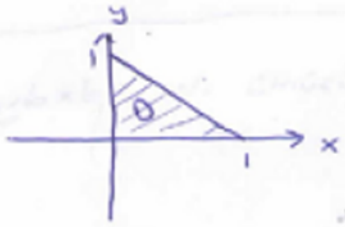
$$A=\iint_{D'}\frac{1}{3u^2v^2}\,du\,dv=\frac{1}{3}\int_1^2\int_1^3\frac{1}{u^2v^2}\,dv\,du$$

---

(18) $\int_0^1\int_0^{1-x}\sqrt{x+y}\cdot(y-2x)^2\,dy\,dx$ integralini $u=x+y$ $v=y-2x$ dönüşümü ile yeniden yazarsak yeni int.?

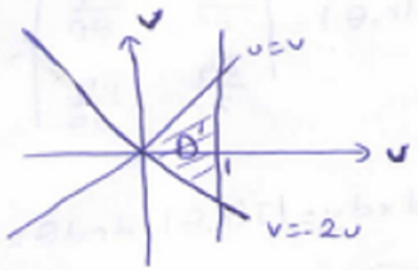
(1)

$D:\begin{cases} x=1 \\ x=0 \\ y=1-x \\ y=0 \end{cases}$



$u=x+y$ $v=y-2x$

D	D'
$x+y=1 \rightarrow$	$u=1$
$x=0 \rightarrow$	$u-v=0 \rightarrow u=v$
$y=0 \rightarrow$	$2u+v=0 \rightarrow v=-2u$



② $u = x+y \quad v = y-2x$

$$J = \frac{1}{\begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}} = \frac{1}{\begin{vmatrix} 1 & 1 \\ -2 & 1 \end{vmatrix}} = \frac{1}{3} \quad \Rightarrow dx\,dy = \frac{1}{3}\,du\,dv$$

③ $$\iint_0 \sqrt{x+y}\cdot(y-2x)^2\,dx\,dy = \iint_{0'} \sqrt{u}\cdot v^2\cdot\frac{1}{3}\,dv\,du$$

$$= \int_0^1 \int_{-2u}^{u} \sqrt{u}\cdot v^2\,\frac{1}{3}\,dv\,du$$

Cevap B

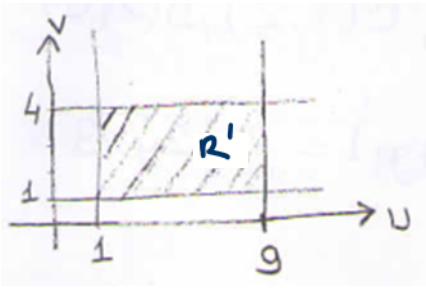
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⑲ $R: \begin{cases} xy=1 \\ xy=9 \\ y=x \\ y=4x \end{cases}$ → 1. Bölgedeki R bölgesinin alanını veren integrali $xy=u$, $\frac{y}{x}=v$ dön. ile yazınız.

$xy=u \qquad \frac{y}{x}=v$

①

R	R'
$xy=1$	$\rightarrow u=1$
$xy=9$	$\rightarrow u=9$
$\frac{y}{x}=1$	$\rightarrow v=1$
$\frac{y}{x}=4$	$\rightarrow v=4$



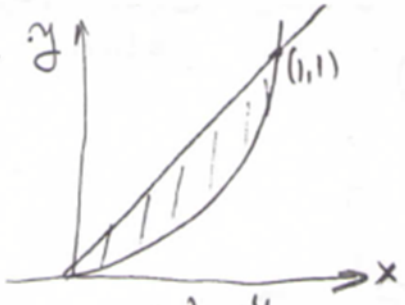
② $$J = \frac{1}{\begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}} = \frac{1}{\begin{vmatrix} y & x \\ -\frac{y}{x^2} & \frac{1}{x} \end{vmatrix}} = \frac{1}{2\frac{y}{x}} = \frac{1}{2v} \rightarrow dx\,dy = \frac{1}{2v}\,du\,dv$$

③ $$A = \iint_R dx\,dy = \int_1^9 \int_1^4 \frac{1}{2v}\,dv\,du$$

Cevap B

(20) R: $y=x$, $y=x^3$, $x\geq 0$, $y\geq 0$ olsun. $f(x,y)=e^{2x^2-x^4}$ fonksiyonunun R bölgesindeki ortalama değeri?

a) e b) e-1 c) e-2 d) e+1



$$\text{Ort. Değer} = \frac{1}{\text{R nin Alanı}} \iint_R f\,dA$$

$$\iint_R dA = \int_0^1 \int_{x^3}^{x} dy\,dx = \int_0^1 \int_y^{\sqrt[3]{y}} dx\,dy = 1/4$$

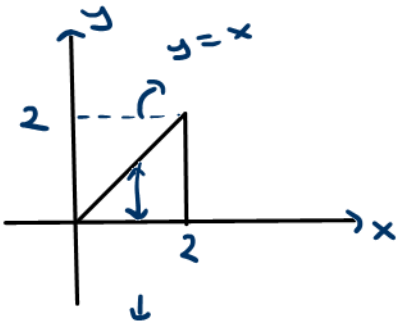
$$\iint_R e^{2x^2-x^4}\,dA = \int_0^1 \int_{x^3}^{x} e^{2x^2-x^4}\,dy\,dx = \int_0^1 e^{2x^2-x^4}(x-x^3)\,dx = \int_0^1 \frac{1}{4} e^u\,du = \frac{1}{4}(e-1)$$

Ortalama Değer $= \frac{1}{1/4} \cdot \frac{1}{4}(e-1) = e-1$

Cevap B

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(21) Köşe noktaları $(0,0)$, $(2,0)$ ve $(2,2)$ olan üçgen bölge üzerinde $f(x,y)=x$ fonksiyonunun ortalama değeri nedir?



$$\bar{f} = \frac{\iint_0 f\,dx\,dy}{\text{0'nin alanı}}$$

0'nin alanı $= \frac{2.2}{2} = 2$

$$\iint_0 f\,dx\,dy = \int_0^2 \int_0^x x\,dy\,dx$$

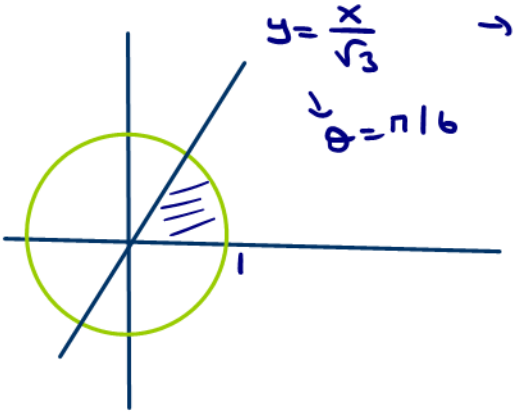
$$= \int_0^2 xy\Big|_0^x dx = \int_0^2 x^2\,dx$$

$$= \frac{x^3}{3}\Big|_0^2 = \frac{8}{3}$$

$$\bar{f} = \frac{\frac{8}{3}}{2} = \frac{4}{3}$$

Cevap A

22 D bölgesi; 1. bölgede, $y = \frac{1}{\sqrt{3}}x$ doğrusu, x-ekseni ve $x^2 + y^2 = 1$ çemberi ile sınırlı bölge olmak üzere D bölgesi üzerinde $\int\int \cos(x^2 + y^2)dA$ integralinin kutupsal formu aşağıdakilerden hangisidir?



$y = \frac{x}{\sqrt{3}} \rightarrow \frac{y}{x} = \frac{1}{\sqrt{3}} \rightarrow \tan\theta = \frac{1}{\sqrt{3}} \rightarrow \theta = \frac{\pi}{6}$

$x = r\cos\theta \quad y = r\sin\theta$

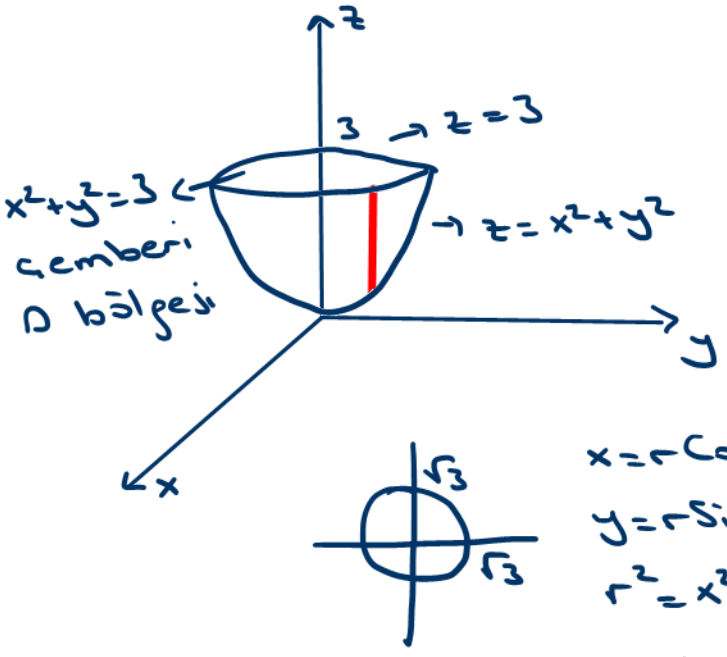
$dxdy = r\,dr\,d\theta \quad x^2 + y^2 = r^2$

$$I = \int_0^{\pi/6}\int_0^1 \cos r^2 \cdot r\,dr\,d\theta$$

Cevap A

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23 $z = x^2 + y^2$ yüzeyi ve $z = 3$ düzlemi arasında kalan hacmi veren integral aşağıdakilerden hangisidir?



$$V = \iint_D (\text{yük.})\,dx\,dy$$

$$= \iint_D (3 - x^2 - y^2)\,dx\,dy$$

$$= \int_0^{2\pi}\int_0^{\sqrt{3}} (3 - r^2)\,r\,dr\,d\theta$$

$x = r\cos\theta$

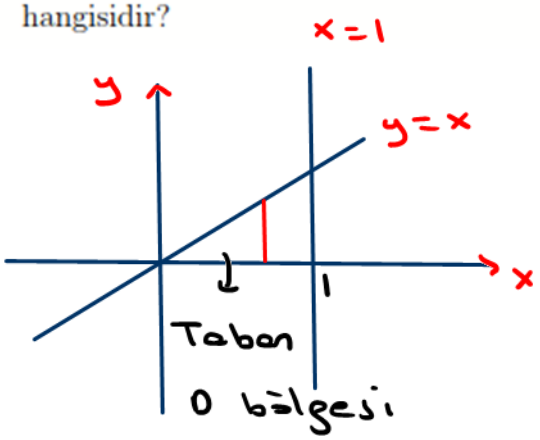
$y = r\sin\theta$

$r^2 = x^2 + y^2$

$dxdy = r\,dr\,d\theta$

Cevap D

24 Alttan xy-düzlemindeki $x \leq 1$, $y \geq 0$, $y \leq x$ eşitsizlikleri ile tanımlanan bölge ve üstten $z = 4 - 4(x^2 + y^2)$ yüzeyi ile sınırlanan katı cismin hacmini veren integral aşağıdakilerden hangisidir?



$$V = \iint_0 (4 - 4(x^2+y^2))\, dx\, dy$$

$$= \int_0^1 \int_0^x (4 - 4(x^2+y^2))\, dy\, dx$$

Cevap A

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25 R bölgesi, $x = 0$, $y = 0$ ve $x - y = 1$ doğruları ile sınırlı bölge olmak üzere $\int\int_R e^{(x-y)(x+y)} dxdy$ integrali, $x - y = u$, $x + y = v$ dönüşümleri altında aşağıdaki integrallerden hangisiyle ifade edilebilir?

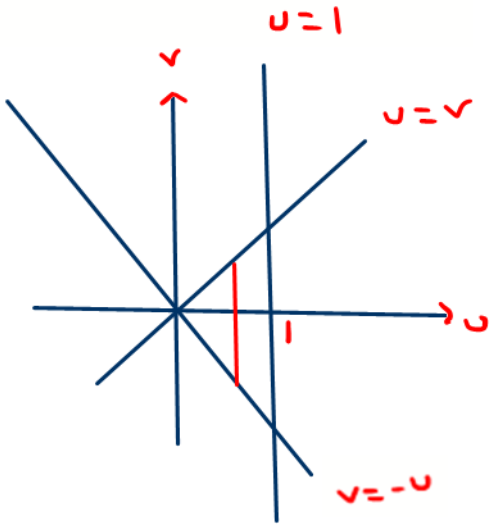
$u = x - y \quad v = x + y$

① $R \rightarrow R'$

$x - y = 1 \rightarrow u = 1$

$x = 0 \rightarrow u = -y \quad v = y \rightarrow u = -v$

$y = 0 \rightarrow u = x \quad v = x \rightarrow u = v$



② $$J = \frac{1}{\begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}} = \frac{1}{\begin{vmatrix} 1 & -1 \\ 1 & 1 \end{vmatrix}} = \frac{1}{2} \Rightarrow dx\,dy = \frac{1}{2} du\,dv$$

③ $$\iint_R e^{\overbrace{(x-y)}^{u}\overbrace{(x+y)}^{v}} dx\,dy = \int_0^1 \int_{-u}^{u} e^{uv} \frac{1}{2} dv\,du$$

Cevap B